

Requirements / Specification Document

TITLE: Power Meters and Components

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
Radio Frequency Technology Division
Guided Wave Electromagnetics Group (672.01)

General Statement of Need:

The Communications Technology Laboratory (CTL) at the National Institute of Standards and Technology (NIST) enables science and industry by providing essential measurement methods, instrumentation, and standards to support all phases of communications technology development, from fundamental characterization and discovery to production.

To support research and measurement services in radio frequency and microwave power, the Radio Frequency Technology Division in CTL requires metering electronics. These items are being acquired as part of an agreement with the Navy Primary Standards Laboratory. The equipment must be new (not refurbished or reconditioned) and covered by the original manufacturer's full warranty with service provided by that manufacturer's U.S.-based service organization.

Item Specifications and Quantities:

Item 1:	Nanovolt Meter Keysight 34420A	
Quantity:	4 Each	
Specifications:	1.1	Each meter shall include option 120: AC line selection 120-volt
Item 2:	RF Power Meter Keysight N432A RF Power Meter	
Quantity:	4 Each	
Specifications:	2.1	Each meter shall include option 008: Thermistor sensor adaptor cable 5 ft (1.5 m)
Item 3:	EPM Series Power Meter Keysight N1913B	
Quantity:	4 Each	

Requirements / Specification Document

Specifications:	3.1	Each meter shall include Option 005: Power sensor cable (x1), 11730A length 5-ft (1.5m)
	3.2	Each meter shall include Option C01: Front calibrator, front sensor
	3.3	Each meter shall include Option 300: 100 - 240 VAC 50/60Hz input frequency
Item 4:	USB Thermocouple Power Sensor Keysight U8481A	
Quantity:	1 Each	
Specifications:	4.1	Each sensor shall include Option 200: DC to 18 GHz
	4.2	Each sensor shall include Option 301: Power sensor cable; 5 ft (1.5 m)
Item 5:	USB Thermocouple Power Sensor Keysight U8487A	
Quantity:	2 Each	
Specifications:	5.1	Each sensor shall include Option 200: DC to 50 GHz
	5.2	Each sensor shall include Option 301: Power sensor cable; 5 ft (1.5 m)
Item 6:	Power Splitter, DC To 18 GHz Keysight 11667A	
Quantity:	4 Each	
Specifications:	6.1	Each splitter shall include Option 002: Type N female input and APC7 output connector
Item 7:	Power Splitter, DC To 50 GHz Keysight 11667C	
Quantity:	2 Each	
Specifications:	7.1	Standard configuration
Item 8:	Slotless Adapter, APC7 to Type-N Female Keysight 85054-60031	
Quantity:	4 Each	
Specifications:	8.1	Standard configuration
Item 9:	Adapter, 2.4 mm Male to 2.4 mm Slotless Female	

Requirements / Specification Document

Keysight 85056-60007		
Quantity:	4 Each	
Specifications:	9.1	Standard configuration
Item 10: Adapter, 2.4 mm Male to 3.5 mm Slotless Female Keysight 11901C		
Quantity:	2 Each	
Specifications:	10.1	Standard configuration
Item 11: Coaxial Attenuator, DC to 18 GHz, Type-N Keysight 8491B		
Quantity:	2 Each	
Specifications:	11.1	Each attenuator shall include option 030: 30 dB attenuation
Item 12: Coaxial Attenuator, DC to 50 GHz Keysight 8490D		
Quantity:	2 Each	
Specifications:	12.1	Each attenuator shall include option 030: 30 dB attenuation

Vendor Requirements

1. Possess at time of award/contracting an Export Control Classification Number (ECCN).
2. Possess at time of award/contracting a Unique Entity Identification Number (UEI) which registration is listed as "Active" on the GSA System for Award Management (SAM.gov).

Procurement Support Information and Ancillary Services:

[Provide input for each section Below or Mark Not Applicable (N/A)]

Section	Requirements/Specifications
Planning Considerations	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain Requirements.]
☒ N/A	☐ US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.)

Requirements / Specification Document

- ☐ Vendor Representative On-Site more than three (3) days.
- ☐ Other Considerations:
[List other considerations to ensure an accurate quote.]

Shipping &
Delivery
☐ N/A

Shipping Address:

- ☒ NIST-Boulder
ATTN: Christian Long
Mail Stop: 672.01
325 Broadway
Boulder, CO 80305

- ☐ NIST-Gaithersburg
ATTN:
Mail Stop:
100 Bureau Dr
Gaithersburg MD 20899
- ☐ [Alternate Shipping Address]

Shipping Criteria:

- ☒ Partial Delivery Acceptable
- ☐ [Enter Shipping and Delivery information by item.]
- ☐ Direct Delivery to OU Building Required
[Explain Need for Requirement]
Building /Room Number: [Enter Building/Room Number]
☐ Building has Loading Dock
Containerization Preference: [Enter Container Type]
Other Requirements:
[List Any Other Requirement]

Delivery Date Criteria:

- ☒ Delivery Shall Be Completed No Later Than:
14 weeks after contract award date
- ☐ Delivery Shall Be Made No Earlier Than:
Click or tap to enter a date.
- ☐ Scheduled Deliveries:

Item Number	Due Date

- ☐ Other:

Requirements / Specification Document

[List Other Shipping, Delivery or Special Requirements.]

Electronic Media

[Enter Electronic Media Requirements by Item.]

☒ N/A

- ☐ Software
- ☐ Electronic Manuals
- ☐ License Key
- ☐ Other:

[List Any Other Electronic Media Requirements]

- ☐ US Government Email Address for Software Delivery:
[Enter Address]

Installation

[Enter Installation Requirements by Item]

☒ N/A

- ☐ Vendor Technician on Site
- ☐ Rigging
- ☐ Uncrating / Unpackaging
- ☐ Removal of Packaging Material
- ☐ Equipment Set Up
- ☐ Start-Up Services
- ☐ Turnkey Installation
- ☐ Other:

[Enter Details]

NOTE: TPOC shall be the coordination POC.

Facility / Utility Considerations

[List Site Specific Utility Considerations by Item]

☒ N/A

- ☐ OFPM Work Order Numbers

[List Work Order Numbers]

- ☐ Power Available

Voltage: [Enter voltage] (V)

Maximum Current: [Enter max current draw] (A)

Phase: [Choose Phase]

Frequency: [Typically 60Hz] (Hz)

NEMA Plug Type: [Enter Plug Type] (i.e.: NEMA L5-20P)

Environment: [Choose Environment]

Other Power Considerations:

Requirements / Specification Document

[List Any Other Power Considerations.]

☐ Utility Connections:

[List Relevant Available Utilities, Pressure(s) and Flowrate(s).]

☐ Other Considerations:

[List Other Considerations to Ensure an Accurate Quote.]

Inspection and
Acceptance

☐ N/A

[Inspection Requirements Per Item (Who, When, What, Where)]

☐ On-Site

☒ Acceptance Not Expected to Exceed 7 Business Days

☐ Acceptance Expected to Exceed 7 Business Days:

[Duration]

☐ Inspection and Acceptance Plan:

[Explanation]

☐ Other:

[Enter Other Requirements]

Training

☒ N/A

Training Requirements (Who, When, What, Where)

☐ Number of Trainees:

[Enter Number of Trainees]

☐ On-Site

☐ Off-Site

Location: [Enter Location(s)]

Is Trainee Travel Required:

[Choose Answer]

Scope of Training

☐ Operation ☐ Maintenance ☐ Troubleshooting

☐ Safety ☐ Other

[Enter Other Scope of Training Requirements]

☐ Training Materials to be Provided by Government:

[Describe Materials]

☐ Training Materials to be Provided by Vendor:

[Describe Materials]

☐ Other

[Enter Other Training Requirements]

NOTE: SME shall be coordination POC.

Requirements / Specification Document

Warranty

☐ N/A

Base Manufacturer Warranty is acceptable

Warranty Length: [i.e. 3 - Years]

What is included?

- ☐ Labor ☐ Parts ☐ Travel
- ☐ Cost and Liability for Returns
- ☐ Other
[Enter Other Requirements]
- ☐ 1 Option Period
- ☐ 2 Option Periods
- ☐ 3 Option Periods
- ☐ 4 Option Periods (Maximum)

Calibration

☒ N/A

[Describe Calibration Requirements Per Item]

Calibration Plan Length: [i.e. 3 - Years]

What is Included?

- ☐ Labor ☐ Spare Parts ☐ Consumables
- ☐ Travel ☐ Cost and Liability for Shipment
- ☐ Software Updates ☐ Manuals
- ☐ Other

[Enter Other Requirements]

☐ Requirements:

[Enter Requirements] (i.e.: On-Site Within 72 hours, etc.)

☐ 1 Option Period

Length:

[i.e. 1 – Year / 6 - Months]

☐ 2 Option Periods

☐ 3 Option Periods

Requirements / Specification Document

☐ 4 Option Periods (Maximum)

Maintenance

☒ N/A

[Describe Maintenance Requirements Per Item]

Maintenance Plan Length: [i.e. 3 - Years]

What is Included?

☐ Labor ☐ Spare Parts ☐ Consumables

☐ Travel ☐ Cost and Liability for Shipment

☐ Software Updates ☐ Manuals

☐ Other

[Enter Other Requirements]

☐ Requirements:

[Enter Requirements] (i.e.: On-Site Within 72 hours, etc.)

☐ 1 Option Period

Length: [i.e. 1 – Year / 6 - Months]

☐ 2 Option Periods

☐ 3 Option Periods

☐ 4 Option Periods (Maximum)

Government Furnished Material / Equipment

☒ N/A

List Any Government Furnished Material / Equipment Per Item]

☐ Samples for Testing

☐ Equipment which will leave Government Site

Description: [Enter Description]

Serial Number: [Enter Serial Number]

NIST Property Number: [Enter NIST Property Number]

☐ Other: [Enter Other Details]

Travel

☒ N/A

[Describe any Travel Requirements Per Item]

☐ Explain:

[Enter Justification / Requirement]

☐ Other:

[Enter Other Requirements]

NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.

Requirements / Specification Document

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